Herding as the Contragredient of Conformal Prediction

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Abstract

Conformal calibration and herding each achieve $O(1/n)$ accuracy by the same finite-sample device: a cumulative discrepancy held bounded by an absolute constant and divided by the sample size. We give the two constructions parallel formal treatments and show that they control opposite arguments of a single bilinear pairing between test functions and measures, $\langle h, \nu \rangle = \int h \, d\nu$: herding steers the empirical measure ν against fixed tests, while conformal calibration steers the acceptance test h — and, in the validity proof, transports the evaluation index — against the fixed empirical measure of scores. The correspondence is stated as a contragredient relation for the permutation action, and it accounts for the linear τ/n dependence penalty of Barber and Pananjady (2026), where blocking arguments yield $\sqrt{\tau/n}$. Since the conformal mechanism (Vovk et al., 2005) predates herding (Welling, 2009), the contragredient is herding.

1 Introduction

Two finite-sample constructions, developed independently, obtain $O(1/n)$ rates where sampling arguments give $O(n^{-1/2})$: the calibration step of conformal prediction, whose mechanism dates to the confidence machines of the late 1990s (see Vovk et al., 2005, for the canonical treatment, and Lei et al., 2018, for the split form used here), and herding (Welling, 2009). This note gives the two mechanisms parallel formal statements (Sections 2 and 3), proves that they are dual actions on the two sides of a single pairing (Section 4), and applies the observation to the analysis of conformal prediction under temporal dependence (Section 5). Interpretive remarks, including the relation to the Bayesian-quadrature reading of Snell and Griffiths (2025), are collected in Section 6.

2 Conformal calibration

Fix a miscoverage level $\alpha \in (0,1)$ and write $p = 1-\alpha$. Let $S_1, \ldots, S_n$ be calibration scores and S_{n+1} a test score, assumed almost surely distinct, and let $k = \lceil p\,(n+1) \rceil$, assumed $\leq n$. Split conformal prediction declares coverage when $S_{n+1} \leq q_n$, where $q_n = S_{(k)}$ is the k-th smallest calibration score.

The choice of q_n is a feedback rule on the acceptance test $h_q(s) = \mathbf{1}\{s \leq q\}$.

Lemma 1 (Steering identity)**.** *With $q_n = S_{(k)}$ and the scores distinct,*

$$\sum_{i=1}^{n} \big(h_{q_n}(S_i) - p \big) \;=\; k - pn \;\in\; [\,p,\; 1+p\,).$$

In particular the cumulative acceptance-rate discrepancy is bounded by an absolute constant independent of n, and the empirical acceptance rate differs from p by at most $(1+p)/n$.

Proof. Exactly k of the n calibration scores satisfy $S_i \le S_{(k)}$, so the sum equals $k - pn$. Since $p(n+1) \le k < p(n+1) + 1$, we have $p \le k - pn < 1 + p$. $\square$

The control-theoretic reading of conformal prediction is established in the online setting: the level update of adaptive conformal inference (Gibbs and Candès, 2021) is integral feedback on realized miscoverage, and Angelopoulos et al. (2023) make the identification explicit as proportional–integral–derivative control. There the controlled quantity is the time-average miscoverage along a sequence; Lemma 1 is the batch counterpart, a single dead-beat correction applied to the calibration set before any test data arrive.

Validity rests on a deterministic counting identity concerning the *placement* of the test index.

Proposition 1 (Counting identity). *Let $s_1, \dots, s_{n+1}$ be distinct reals and, for $i \in \{1, \dots, n+1\}$, call placement i* covered *if s_i is at most the k-th smallest of $\{s_j\}_{j \ne i}$. Then placement i is covered if and only if the rank of s_i among all $n+1$ values is at most k, and consequently exactly k of the $n+1$ placements are covered.*

Proof. Placement i is covered iff at most $k-1$ of the other n values are smaller than s_i, iff the rank of s_i among all $n+1$ values is at most k. The ranks are a permutation of $\{1, \dots, n+1\}$, so exactly k indices qualify. $\square$

Corollary 1 (Marginal validity). *If $(S_1, \dots, S_{n+1})$ is exchangeable with almost surely distinct components, then $\mathbb{P}\{S_{n+1} \le q_n\} = k/(n+1) \ge 1 - \alpha$.*

Proof. The coverage event $\{S_{n+1} \le q_n\}$ is the event that placement $n+1$ is covered, since q_n is the k-th smallest of the values other than S_{n+1}. By exchangeability the indicator of "placement i is covered" has the same expectation for every i; average Proposition 1 over i. $\square$

No probabilistic inequality enters before the final step, and the final step is an identity in distribution rather than a concentration bound.

3 Herding

Let $\phi : \mathcal{X} \to \mathcal{H}$ be a feature map into a Hilbert space and $\mu = \mathbb{E}_P\, \phi(X)$ a target mean. Herding (Welling, 2009) generates points by the recursion

$$x_t \;=\; \arg\max_{x \in \mathcal{X}} \langle w_{t-1}, \phi(x) \rangle, \qquad w_t \;=\; w_{t-1} + \mu - \phi(x_t), \tag{1}$$

with w_0 given.

Proposition 2 (Herding discrepancy). *For any sequence produced by* (1),

$$\Big\| \frac{1}{n} \sum_{t=1}^{n} \phi(x_t) - \mu \Big\| \;=\; \frac{\|w_0 - w_n\|}{n}.$$

In particular, if $\sup_t \|w_t\| \le B$ then the feature-mean discrepancy is at most $2B/n$.

Proof. Telescoping (1) gives $w_n = w_0 + n\mu - \sum_{t \le n} \phi(x_t)$; rearrange and divide by n. $\square$

Conditions under which the weights remain bounded are given by Welling (2009) in finite dimension and analysed in general via the identification of (1) with the Frank–Wolfe algorithm on the marginal polytope (Bach et al., 2012); Chen et al. (2010) treat the kernel case. Under such conditions herding attains $O(1/n)$ where i.i.d. sampling attains $O(n^{-1/2})$.

4 The duality

Sections 2 and 3 exhibit the same ledger: a cumulative discrepancy — $k - pn$ in Lemma 1, $w_0 - w_n$ in Proposition 2 — bounded by a constant and divided by n. To compare the mechanisms, write both in terms of the bilinear pairing between bounded test functions h and finite measures ν,

$$\langle h, \nu \rangle \;=\; \int h \, d\nu.$$

Each mechanism controls one argument of this pairing and holds the other fixed. Herding fixes the family of tests (the coordinates of ϕ, whose ν-averages are to match μ) and constructs the measure, $\nu_n = \frac{1}{n}\sum_t \delta_{x_t}$. Conformal calibration fixes the measure — the empirical distribution of scores — and constructs the test h_{q_n}; its validity proof additionally moves the evaluation index, as follows.

Let ρ denote the permutation representation of the symmetric group $\mathfrak{S}_{n+1}$ on $\mathbb{R}^{n+1}$, $\rho(\pi)e_i = e_{\pi(i)}$, and let $\rho^*(\pi) = \rho(\pi^{-1})^\top$ be the contragredient representation on the dual space, so that $\rho^*(\pi)e_j^* = e_{\pi(j)}^*$ and $\langle \rho^*(\pi)f, \rho(\pi)v\rangle = \langle f, v\rangle$.

Proposition 3 (Two descriptions of the orbit average)**.** *For $s \in \mathbb{R}^{n+1}$ with distinct entries let $\chi_i(s) = \mathbf{1}\{s_i \le Q_k(s_{-i})\}$, where $Q_k(s_{-i})$ is the k-th smallest entry of s with coordinate i deleted. Then for every $\pi \in \mathfrak{S}_{n+1}$,*

$$\chi_{n+1}\big(\rho(\pi)s\big) \;=\; \chi_{\pi^{-1}(n+1)}(s).$$

Consequently the average of χ_{n+1} over rearrangements of the data equals the average of $s \mapsto \chi_i(s)$ over placements i of the evaluation index — equivalently, over the orbit $\{\rho^*(\pi)\, e_{n+1}^*\}$ of the evaluation functional at fixed data — and Corollary 1 may be proved on either side of the pairing.

Proof. Q_k is invariant under permutations of its n arguments, and coordinate $n+1$ of $\rho(\pi)s$ is $s_{\pi^{-1}(n+1)}$; substitute. □

Remark 1 (Direction of the duality). Permutation representations are orthogonal, so ρ^* is equivalent to ρ; the content of the contragredient formulation is not a new representation but the location of the action — on the covector (test) side of the pairing for conformal prediction, on the vector (measure) side for herding. Chronologically the index-side mechanism came first: the conformal construction predates herding by roughly a decade. In that sense it is herding that realises the contragredient of conformal prediction, in the primal space of generated points, with active selection of x_t replacing the transport of an evaluation index.

The correspondence is summarised below.

	Conformal calibration	**Herding**
controlled side	test h_{q_n} (and evaluation index)	measure ν_n
fixed side	empirical score measure	feature tests ϕ
ledger	$k - pn \in [p, 1+p)$	$w_0 - w_n$, $\lVert w_t \rVert \le B$
rate	$(1+p)/n$	$2B/n$
probabilistic input	exchangeability	none (deterministic target)

Remark 2 (An image). In a gambling idiom: herding moves the balls to balance a fixed wheel; conformal prediction leaves the balls where they fell and relabels the slots under the pointer.

5 Dependent data

The dual reading extends to the time-series analysis of Barber and Pananjady (2026), where exchangeability fails and the transport step of Proposition 3 is no longer distribution-preserving. Their argument deletes a buffer of τ coordinates before applying the transport. Two facts price this operation. First, deleting τ of n entries moves the k-th order statistic by at most τ ranks, so the quantile level shifts by at most $\tau/(n+1)$ — Lemma 1 degrades linearly in the number of damaged slots. Second, the residual failure of transport invariance is charged at its total-variation price, the switch coefficient $\bar{\Psi}_\tau$, bounded by $2\beta(\tau)$ for a stationary β-mixing sequence. The resulting floor,

$$\mathbb{P}(\text{cover}) \ \geq \ 1 - \alpha - \min_{\tau} \Big\{ \tfrac{\tau}{n+1} + \bar{\Psi}_\tau \Big\},$$

contains no concentration step, and correspondingly no square root: when the same problem is attacked by blocking and empirical-process methods (Oliveira et al., 2024) the rate degrades to $\sqrt{\tau/n}$. The advantage of Barber and Pananjady (2026) over blocking is the advantage of a bounded ledger over sampling error — Proposition 2's advantage over Monte Carlo, transported across the pairing.

6 Interpretation and limits

Remark 3 (Quadrature). Snell and Griffiths (2025) recast the conformal computation as Bayesian quadrature over the calibration scores. That is a measure-side reading of the same pairing: quadrature weights on scores on one side, a transported evaluation functional on the other.

Remark 4 (Limits). The duality concerns the finite-sample ledger only. The passage from Proposition 1 to Corollary 1 is exchangeability's, or, in Section 5, its mixing surrogate's; no bookkeeping supplies it. Nor does control of thresholds and labels create information: it re-levels coverage while conditional sharpness remains a property of the scores (Cotton). Whether an actively *aimed* index-side controller exists — selecting or reweighting calibration slots to balance the rank count conditionally, as herding aims its particles — is open; localized and weighted conformal methods are the nearest existing steps, and the same information-theoretic ceiling applies to them.

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